Lund University
Mathematical Sciences
Mathematical Statistics

HT2025

# Computer Lab 2
# Probability Theory

## 1 Introduction

The purpose of this computer lab is to illustrate some important concepts in probability theory. The computer lab consists of two parts.

1. The law of large numbers (LLN)
2. The central limit theorem (CLT)

Try to give answers to all questions. Do not continue until you have understood a part/moment, and ask when you do not understand.

## 2 The law of large numbers (LLN)

The law of large numbers is arguably the most important result in mathematical statistics. One version of it says that if $X_1, X_2, \ldots$ is a sequence of independent identically distributed r.v.'s with expectation $m$, then for every interval $(m-\epsilon, m+\epsilon)$ with $\epsilon > 0$, the r.v. $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$ lies in this interval with high probability, if $n$ is large enough. This part of the lab consists of illustrating the LLN.

1. Generate 10000 $Un(0,1)$ distributed r.v.'s and plot the sequence of averages $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$ against $n$.

```
n<-10000
x<-runif(n)
x_n<-cumsum(x)/(1:n)
plot(1:n,x_n)
```

   Q1 What does the graph seem to converge to? Calculate (analytically) $E(X_1)$.

   Q2 How does this relate to the statement of the LLN?

2. Generate 10000 $N(0,1)$-distributed r.v.'s and form $Z_n = n^{-1}\sum_{i=1}^{n} X_i^2$, and plot $Z_n$ versus $n$.

```
n<-10000
x<-rnorm(n,0,1)
z_n<-cumsum(x^2)/(1:n)
plot((1:n),z_n)
```

   Q1 What does the graph seem to converge to? Calculate $E(X_1^2)$.

   Q2 How does this relate to the statement of the LLN.

Q3 Explain how this can be used to give approximations for $E(g(X))$ for functions $g$ (see also part 4 below.) For which functions $g$ can this be done? Give a restriction on $g$ that ensures that this can be done.

3. Simulate 10000 coin tosses, with

$$X_i = \begin{cases} 0 \text{ if tail up in toss number } i, \\ 1 \text{ if head up in toss number } i, \end{cases}$$

for $i = 1, \ldots, 10000$. The model is that $P(\text{head}) = 0.39$ och $P(\text{tail}) = 0.61$. Form $X_n = n^{-1} \sum_{i=1}^{n} X_i$ and plot versus $n$.

```
n<-10000
x<-runif(n)
x_i<-as.double(x<0.39)
z_n<-cumsum(x_i)/(1:n)
plot((1:n),z_n)
```

Q1 What does the graph converge to? Calculate (analytically) $E(X_1)$.

Q2 How does the result relate to what the LLN says?

Q3 Explain how this can be used to get approximations to $P(A)$ for arbitrary events $A$.

4. Let us study the integral $I = \int_0^1 e^{-x^2}\, dx$. There is no closed form expression for $I$, since we lack an elementary expression for the anti-derivative of $e^{-x^2}$. We will do a so called Monte-Carlo approximation of $I$. We can write $I = E(g(X))$, with $X \in Un(0, 1)$ and $g(x) = e^{-x^2}$. Run the following commands.

```
n<-10000
x<-runif(n)
y<-exp(-x^2)
z_n<-cumsum(x)/(1:n)
plot((1:n),z_n)
```

Q1 What numerical value does the graph seem to converge to?

Q2 Use the LLN to motivate the approximation to $I$.

# 3 The central limit theorem (CLT)

The central limit theorem says that if $X_1, X_2, \ldots$ is a sequence of independent identically distributed r.v.'s then their sum $S_n = \sum_{i=1}^{n} X_i$ and their average $\bar{X}_n = n^{-1} S_n$ are approximately distributed as a Gaussian (or normal) r.v. if $n$ is large. This second part of the lab consist of illustrating the CLT.

In order to do so we will use an estimate $F_n$ of the d.f. $F$ of $X$ (which you will study in great detail in the inference theory part of the course) called the empirical distribution function, defined as

$$F_n(x) = \frac{1}{n} \sum_{i=1}^{n} 1\{X_i \leq x\}.$$

Q1 Derive an analytic expression for $E(F_n(x))$.

Q2 Use the LLN to motivate why $F_n$ can be used as approximation of $F$.

Q3 For a fixed $x_0$: In what sense does $F_n(x_0)$ converge to $F(x_0)$ when $n \to \infty$?

In order to check if the data $x_1, \dots, x_n$ come from a certain distribution $F$ one may estimate $F$ with $F_n$ and compare the two functions $F_n$ (the estimate) and $F$ (what one estimates, the theoretical statement). If $F_n$ and $F$ look alike then it seems reasonable to draw the conclusion that the data $x_1, \dots, x_n$ in fact come from the distribution $F$. We note that the above reasoning in Q2, Q3 explains that $F_n$ is close to $F$ when $n$ is large. This idea is formalised in the second part of the course in inference theory where on can make a formal *test* for the hypothesis that the data come from $F$.

For now we will merely use a visual inspection: if the graphs of $F_n$ and $F$ are close we may draw the conclusion that the data in fact come from $F$. We note however that $F_n$ and $F$ are usually not linear and are thus often curved, and then the visual inspection consists of seeing if two curved functions seem to be the same. Our visual system is not good at seeing such deviations and it is easier for us to see a deviation from linearity. For this reason one may instead plot the function $F_n$ in a transformed coordinate system, which is transformed in such a way the plotting $F$ would give a straight line. Let us see how this may be done.

Let $x_1, \dots, x_n$ be the data and assume for simplicity that we have sorted them so that $x_1 \leq \dots \leq x_n$. Let $q_1 = 1 - F(x_1), q_2 = 1 - F(x_2), \dots, q_n = 1 - F(x_n)$ so that we have $q_i = 1 - F(x_i) = P(X > x_i)$ for $i = 1, \dots, n$. The values $q_i$ are then the *survival probabilities* at $x_i$ and the values $x_i$ are the *theoretical* $q_i$*-quantiles*. Recall the definition of an $q$-quantile for a continues r.v. $X$, it is the solution $x_q$ to the equation

$$q \quad = \quad P(X > x_q) = 1 - F(x_q) \tag{1}$$

Now let us replace $F$ with $F_n$ and do an analogous reasoning. We have that $q_1^{(n)} = 1 - F_n(x_1), \dots, q_n^{(n)} = 1 - F_n(x_n)$, so that $q_i^{(n)} = 1 - F_n(x_i)$ for $i = 1, \dots, n$. But note now that by the definition of $F_n$ in fact $q_i^{(n)} = 1 - i/n$.

Now we see that if $F_n$ is close to $F$, then also $q_i$ must be close to $q_i^{(n)}$. Visually we can check for this by plotting the points $(q_i, q_i^{(n)}), i = 1, \dots, n$, i.e. the points $(q_i, 1 - i/n), i = 1, \dots, n$ in a coordinate system. Such a plot is called a PP-plot. Then if $F_n$ is close to $F$ the points $(q_i, 1 - i/n), i = 1, \dots, n$ will be close to the straight line $y = x$ in the coordinate system. Thus we may look for deviation from a straight line in order to assess that the data $x_1, \dots, x_n$ do not come from the distribution $F$.

An alternative to the PP-plot is the QQ-plot. In that we plot the empirical quantiles versus the theoretical quantiles as follows: The empirical quantiles are the data $x_1 \dots, x_n$. They correspond to the empirical survival probabilities $1 - F_n(x_i) = 1 - i/n$, and are the solutions to these equations. Now if $F_n$ is close to $F$ then $1 - F(x_i)$ is close to $1 - F_n(x_i) = 1 - i/n$. The theoretical quantiles (in fact the theoretical $(1 - i/n)$-quantiles as defined in (1) above) are the solutions $\tilde{x}_i$ to the equations $1 - F(\tilde{x}_i) = 1 - i/n$, i.e. to the equations

$$F(\tilde{x}_i) \quad = \quad i/n.$$

Then if $F_n$ is close to $F$ we will have that $x_i$ are close to $\tilde{x}_i$, and then the points $(x_i, \tilde{x}_i)$ will be close to the straight line $y = x$ in the cartesian coordinate system. The QQ-plot is a plot of those points[1], and a deviation from a straight line of the points indicates that the data do not come from the distribution $F$,.

A final tweak of this idea/approach is when the distribution $F$ that one wants to check for depends on unknown parameters. An example is when one wants to check if data come from a Gaussian or normal distribution, but where one does not know the expectation and variance. Thus the distribution function one wants to check for is

$$\begin{aligned} F(x) &= F_{\mu,\sigma^2}(x) \\ &= \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(t-\mu)^2/2\sigma^2} dt, \end{aligned}$$

[1]The choice of exactly the points $i/n$ is not a must, and in fact there are many other choices possible, this is sometimes called a choice plotting positions

and $\mu, \sigma^2$ are unknown. An approach is then to *estimate* the unknown parameters from the data, for instance by

$$\begin{aligned}\hat{\mu} &= \frac{1}{n}\sum_{i=1}^{n} x_i, \\ \hat{\sigma}^2 &= \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \hat{\mu})^2,\end{aligned}$$

where the "hat" notations is to indicate that the values estimates. Then we may. use $F_{\hat{\mu},\hat{\sigma}^2}$ as the distribution to check the the data for, and thus we obtain the theoretical quantiles with the use of (1) with $F$ replaced by $F_{\hat{\mu},\hat{\sigma}^2}$.

Do help on “rnorm,rexp,runif,rbinom,qqnorm” and on "apply" before you continue.

1. Generate $n = 100$ normal distributed r.v.'s with expectation $\mu = 2$ and variance $\sigma^2 = 1$. Do a normal probability plot of these.

   ```
   n<-100
   mu<-2
   s2<-1
   x<-rnorm(n,mu,s2)
   qqnorm(x)
   qqline(x)
   ```

   Q1 Do the above with values $n = 10, 20, 200, 1000$. What is the outcome?

2. Generate $n = 100$ exponentially distributed r.v.'s with expectation 1 and form their average $\bar{X}_n$. Do this $m = 200$ times so that you get 200 averages $\bar{X}_n^1, \ldots, \bar{X}_n^{200}$. Do an normal probability plot of these.

   ```
   n<-100
   m<-200
   x<-rexp(n*m)
   x2<-matrix(x,m,n)
   x_n<-apply(x2,1,mean)
   qqnorm(x_n)
   qqline(x_n)
   ```

   Q1 Do the above with values $n = 10, 20, 200, 1000$. What is the outcome?

   Q2 Do the above with varying $m$. What is the outcome?

   Q3 Explain how your results relate to the CLT. Explain what happens qualitatively as $n \to \infty$ and as $m \to \infty$.

3. Do the above with $Un(0, 1)$-distributed r.v.'s and varying $n = 5, 10, 50$.

   ```
   n<-100
   m<-200
   x<-runif(n*m)
   x2<-matrix(x,m,n)
   x_n<-apply(x2,1,mean)
   qqnorm(x_n)
   qqline(x_n)
   ```

   Answer Q1, Q2 and Q3 as above for this case.

   Q4 What, if any, is the difference with the previous exercise?

4. Generate $n = 100$ $Bin(nn, pp)$-distributed r.v.'s with $nn = 3$ and $pp = 0.5$ and their average $\bar{X}_n$. Do this $m = 200$ times. Do a normal probability plot on these.

```
n<-100
m<-200
nn<-3
pp<-0.5
x<-rbinom(n*m,nn,pp)
x2<-matrix(x,m,n)
x_n<-apply(x2,1,mean)
qqnorm(x_n)
qqline(x_n)
```

   Q1 Vary $n = 10, 50, 200$. What is the result?

   Q2 What happens if you change to $nn = 100$, and vary $n = 10, 50, 200$?

   Q3 Can you use the CLT to explain the qualitative difference between the distributions $Bin(3, 0.5)$ and $Bin(100, 0.5)$?

5. Try to imagine what can happen if you change the binomial distribution in the previous exercise to $Bin(100, 0.1)$. Think first, and then do a run and answer Q1 and Q2 in the previous exercise.